

Venkata Krishna Varun

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WORK EXPERIENCE

ORACLE, *Software Development Engineer (Cloud & AI Infrastructure)*

Jul 2025 - Present

- Architected an Agentic AI-based Root Cause Analysis (RCA) system leveraging LLMs, RAG pipelines, and vector databases for autonomous cloud diagnostics
- Automated distributed log analysis and failure localization workflows, reducing manual debugging effort by **30%** across internal infrastructure systems
- Engineered scalable GenAI tooling for contextual retrieval of enterprise operational knowledge across large-scale developer environments
- Developed infrastructure automation utilities and monitoring workflows to improve reliability and operational visibility for cloud-native services

SIMPL (BUY NOW PAY LATER), *Data Engineering Intern*

Jun 2024 - Jul 2024

- Implemented a real-time Change Data Capture (CDC) pipeline replacing legacy daily snapshot-based synchronization workflows
- Deployed Debezium using Kafka Connect and configured distributed PostgreSQL source connectors for continuous row-level event streaming
- Engineered scalable event ingestion pipelines using Apache Kafka, enabling low-latency synchronization across distributed databases
- Developed a high-throughput PySpark parsing framework to process Debezium-generated JSON mutation streams for downstream analytics systems

RESEARCH PROJECTS

REWARD DECOMPOSITION IN MARL-LOMAQ, *Supervised by Dr. Lakshmanan Kailasam*

- Developed a scalable Multi-Agent Reinforcement Learning (MARL) framework using the LOMAQ algorithm under the CTDE paradigm
- Implemented decomposable reward functions to address structural credit assignment challenges in cooperative multi-agent environments
- Applied monotonic value decomposition constraints to align decentralized policies with globally optimal decision-making
- Evaluated convergence stability and reward optimization performance under varying agent coordination settings

SHORT-TERM RESIDENTIAL LOAD FORECASTING USING DEEP LEARNING, *Supervised by Prof. S.K. Singh & Ph.D. Jaishankara*

- Designed and trained recurrent deep learning models using LSTM and GRU architectures for short-term residential energy forecasting
- Achieved an R^2 **score of 0.938** and RMSE of **0.263**, outperforming traditional regression-based forecasting approaches
- Integrated weather-dependent temporal features and anomaly-aware preprocessing using DB-SCAN clustering
- Applied dropout regularization and Adam optimization to improve model generalization under volatile energy consumption patterns

COURSE PROJECTS

TURBO COMPRESSOR, *Under Prof. Dr. Bhaskar Biswas*

- Developed a lossless file compression and decompression CLI utility in **C++** using the Huffman Encoding algorithm for optimized binary data storage
- Implemented priority queues, binary trees, and custom bit manipulation techniques to efficiently encode and decode text streams
- Designed the system to reconstruct original files with zero information loss while minimizing storage overhead
- Applied concepts from Data Structures, Object-Oriented Programming, and Algorithm Design to improve runtime efficiency and modularity

RENT-A-BIKE, *Under Prof. Dr. Ravindranath Chowdary*

- Developed a full-stack bike reservation platform using **Spring Boot**, enabling users to search, reserve, and manage bike bookings through a responsive web interface
- Implemented booking history tracking and invoice generation modules to improve user-side transaction management
- Designed administrator functionalities for bike inventory management, dynamic pricing updates, and centralized booking monitoring
- Integrated relational database operations using **MySQL** and followed MVC-based backend architecture for scalable service management

BLOG BOOK, *Under Prof. Dr. Amrita Chaturvedi*

- Engineered a feature-rich blogging platform using **Django** with secure authentication and role-based content management
- Implemented core functionalities including blog creation, editing, deletion, threaded comments, likes, and subscription workflows
- Developed an admin dashboard for centralized moderation and efficient blog lifecycle management
- Designed scalable backend routing and ORM-based database interactions using Python and Django framework components

EDUCATION

INDIAN INSTITUTE OF TECHNOLOGY (BHU), VARANASI
Bachelor of Technology in Computer Science and Engineering
CPI: 8.46 / 10.00

Jul 2021 - May 2025

SRI CHAITANYA COLLEGE - GRADE 11 & 12
Andhra Pradesh State Board of Intermediate Education
Percentage : 96.6%

Jun 2019 - Jul 2021

ACADEMIC ACHIEVEMENTS & NATIONAL RANKS

- **JEE Advanced 2021:** Secured **All India Rank 1353**, placing in the top **0.9%** among more than **150,000** elite candidates nationwide
- **JEE Main 2021:** Secured **All India Rank 2435**, placing in the top **0.24%** among more than **1 million** applicants nationwide
- **Quantitative Excellence:** Achieved a perfect **100% score in Mathematics** in JEE Main 2021, demonstrating strong analytical and problem-solving aptitude
- **Indian Olympiad in Chemistry (ACT):** Recognized as the **Andhra Pradesh State Topper**
- **Academic Standing:** Graduated with a **Cumulative Performance Index (CPI) of 8.46/10** in Computer Science and Engineering from **IIT (BHU) Varanasi** (Class of 2025)

- **Coursework Excellence:** Secured the highest **Grade of Excellence (A)*** in ITW1 and earned **A Grades** in core subjects including Artificial Intelligence, Data Structures, and Computer Programming

LEADERSHIP

CHAMPIONSHIP WINNER | ORACLE FY26 GLOBAL TECH PROGRAM, *February 2026*

- Spearheaded a **high-performing, 5-member team** ("Think Tank") to architect and pitch high-impact technical solutions in a competitive, global corporate hackathon setting
- Successfully outpaced competing teams under strict delivery timelines to capture the Championship Trophy

FMC WEEKEND, IIT (BHU) VARANASI, *Publicity Executive*

- Led a cross-functional team of **25 members** to execute publicity and outreach operations for the institute's flagship media fest
- Coordinated external outreach campaigns across **200+ colleges** to increase nationwide participation
- Contributed to digital engagement strategies resulting in **100K+ organic impressions** across social platforms

MENTORING

TELUGU CSE COMMUNITY, IIT (BHU), *Lead Mentor*

- Mentored a cohort of **20+ students** in DSA, internship preparation, resume building, and placement strategy
- Conducted mock interviews and structured problem-solving sessions focused on software engineering recruitment
- Designed roadmap-driven preparation plans for internships and full-time technical roles

GAUTH EXPERT, *Subject Matter Expert — Advanced Mathematics*

- Solved **500+** advanced mathematics problems involving Calculus, Algebra, Probability and Coordinate Geometry
- Delivered structured, step-by-step analytical explanations under strict response timelines

SPORTS & EXTRA-CURRICULAR ACTIVITIES

- Won the **Gold Medal in Boxing** at **Spardha 2022**, the annual inter-collegiate sports festival of IIT (BHU) Varanasi
- Successfully completed a **10 KM Marathon** organized for Cancer Awareness, demonstrating endurance, discipline, and commitment toward social causes

RELEVANT COURSES

- Data Structures • Computer Programming • Object Oriented Programming • Operating Systems • Computer Networks • Database Management Systems • Artificial Intelligence • Digital Logic and Design • Computer Graphics • Computer Architecture • Probability and Statistics • Discrete Mathematics • Software Engineering • Distributed Systems • Machine Learning Fundamentals